

Problem statement: -1

Title: Recovery of Organic Solvents from Saline and Acidic Pharmaceutical Effluent

Pharmaceutical manufacturing extensively uses organic solvents in synthesis, extraction, purification, and cleaning operations. A substantial fraction of these solvents is discharged into aqueous effluent streams during production and downstream processing. These effluents typically contain a complex mixture of residual organic solvents, dissolved inorganic salts (from neutralization, salting-out, and pH adjustment steps), and free mineral or organic acids. Such mixed effluents pose serious challenges for waste treatment, environmental compliance, and resource efficiency.

Currently, most pharmaceutical plants treat solvent-laden effluents through conventional effluent treatment plants (ETPs) or dispose of them via off-site hazardous waste handling. High salt concentration inhibits biological treatment efficiency, while residual solvents increase COD and VOC emissions. Acids further complicate treatment by causing corrosion, pH instability, and safety risks. As a result, valuable solvents are lost, treatment costs increase, and achieving sustainability targets becomes challenging.

Problem Statement

Develop a technically feasible, safe, and economically viable process to selectively recover reusable organic solvent(s) from pharmaceutical effluent streams that contain high dissolved salts and residual acids, while ensuring that salts and acids are effectively retained in the aqueous phase.

Cash Prize: - 3.5 k

Time line

1. Submissions of the Final report and presentations of IDP on 10/01/2025)

On mail id ts.ta@stumarj.ictmumbai.edu.in, before 22:00, the subject of the mail should be "Problem statement" final report submission by "team name"

Example: - **Recovery of Organic Solvents from Saline and Acidic Pharmaceutical Effluent, final report submission by Team AAKRITI**